

Problem

Three loads are connected in parallel to a $120\angle 0^\circ \text{ V}$ rms source. Load 1 absorbs 60 kVAR at $\text{pf} = 0.85$ lagging, load 2 absorbs 90 kW and 50 kVAR leading, and load 3 absorbs 100 kW at $\text{pf} = 1$. (a) Find the equivalent impedance. (b) Calculate the power factor of the parallel combination. (c) Determine the current supplied by the source.

Step-by-step solution

Step 1 of 7

(a)

The reactive power absorbed by the first load is,

$$Q_1 = 60 \text{ kVAR}$$

The power factor of first load is,

$$\cos \theta_1 = 0.85 \text{ lagging}$$

Calculate the power angle for the first load.

$$\cos \theta_1 = 0.85 \text{ lagging}$$

$$\begin{aligned}\theta_1 &= \cos^{-1}(0.85) \\ &= 31.79^\circ\end{aligned}$$

Calculate the value of apparent power for the first load.

$$Q_1 = S_1 \sin \theta_1$$

$$\begin{aligned}S_1 &= \frac{Q_1}{\sin \theta_1} \\ &= \frac{60 \text{ kVAR}}{\sin(31.79^\circ)} \\ &= 113.89 \text{ kW}\end{aligned}$$

Step 2 of 7

Calculate the value of real power absorbed by the first load.

$$\begin{aligned}P_1 &= S_1 \cos \theta_1 \\ &= (113.89 \text{ kW})(0.85) \\ &= 96.8 \text{ kW}\end{aligned}$$

Calculate the value of complex power absorbed by the first load.

$$\begin{aligned}\mathbf{S}_1 &= P_1 + jQ_1 \\ &= 96.8 \text{ kW} + j60 \text{ kVAR}\end{aligned}$$

Step 3 of 7

The real power of second load is,

$$P_2 = 90 \text{ kW}$$

The power factor is leading.

For leading power factor $Q < 0$.

The reactive power absorbed by the second load is,

$$Q_2 = -50 \text{ kVAR}$$

Calculate the value of complex power.

$$\begin{aligned} \mathbf{S}_2 &= P_2 + jQ_2 \\ &= 90 \text{ k} - j50 \text{ kVAR} \end{aligned}$$

Step 4 of 7

The power absorbed by the third load is,

$$P_3 = 100 \text{ kW}.$$

The power factor of third load is, $\cos \theta_3 = 1$

Calculate the power angle for first load.

$$\cos \theta_3 = 1 \text{ lagging}$$

$$\begin{aligned} \theta_1 &= \cos^{-1}(1) \\ &= 0^\circ \end{aligned}$$

Calculate the value of reactive power.

$$Q_1 = S_1 \sin \theta_1$$

$$\begin{aligned} Q_3 &= S_1 \sin(0^\circ) \\ &= 0 \end{aligned}$$

Calculate the value of complex power.

$$\begin{aligned} \mathbf{S}_3 &= P_3 + jQ_3 \\ &= 100 \text{ kW} + j(0) \\ &= 100 \text{ kVA} \end{aligned}$$

Step 5 of 7

Calculate the value of total complex power.

$$\begin{aligned} \mathbf{S} &= \mathbf{S}_1 + \mathbf{S}_2 + \mathbf{S}_3 \\ &= [(96.8 + j60) + (90 - j50) + (100)] 10^3 \\ &= (286.8 + j10) \text{ kVA} \\ &= 286.97 \angle 1.996^\circ \text{ kVA} \end{aligned}$$

Calculate the value of equivalent impedance.

$$\mathbf{Z}^* = \frac{(V_{\text{rms}})^2}{\mathbf{S}}$$

Substitute 120 V for V_{rms} , $286.97 \angle 1.996^\circ \text{ kVA}$ for $\mathbf{S}$.

$$\begin{aligned} \mathbf{Z}^* &= \frac{(120)^2}{286.97 \times 10^{-3} \angle 1.996^\circ} \\ &= \frac{14,400}{286.97 \times 10^{-3} \angle 1.996^\circ} \\ \mathbf{Z}^* &= 50.14 - j1.7509 \text{ m}\Omega \\ \mathbf{Z} &= 50.14 + j1.7509 \text{ m}\Omega \end{aligned}$$

Therefore, the value of equivalent impedance is $0.0501 \angle 1.996^\circ \Omega$.

Step 6 of 7

(b)

The total load in polar form is,

$$\mathbf{Z} = 0.0501 \angle 1.999^\circ \text{ m}\Omega$$

Calculate the power factor of parallel combination.

$$\begin{aligned} \cos \theta &= \cos(1.999^\circ) \\ &= \cos(1.996^\circ) \\ &= 0.9994 \end{aligned}$$

Therefore, the power factor of parallel combination is 0.9994 .

Step 7 of 7

(c)

Calculate the value of current supplied by the source.

$$\begin{aligned} I_{\text{rms}} &= \frac{V_{\text{rms}}}{Z^*} \\ &= \frac{120}{0.0501 \times 10^{-3} \angle 1.996^\circ} \\ &= 2.34 \angle -1.996^\circ \text{ kA} \end{aligned}$$

Therefore, the value of current supplied by the source is $2.34 \angle -1.996^\circ \text{ kA}$.